Angle of Arrival Estimation Using Cholesky Decomposition

Saleh O. Al-Jazzar

Abstract

In this paper, an angle of arrival (AOA) estimator is presented. Many applications require accurate AOA estimates such as wireless positioning and signal enhancement using space processing techniques. The proposed AOA estimator depends on the Cholesky decomposition of the received signal autocorrelation matrix. The resultant decomposed matrices are used to modify the crosscorrelation matrix of the received signals at the antenna array doublets. The proposed method is named the Cholesky-decomposition-based-AOA (CDBA) estimator. In comparison with the ESPRIT algorithm which utilizes the eigenvalue decomposition (EVD) of the received signal autocorrelation and crosscorrelation matrices, the decomposition in the CDBA method is performed on a smaller matrix dimension than that in the ESPRIT algorithm. The other advantage is that the CDBA method has better performance than the ESPRIT algorithm. Simulations of the proposed CDBA method are shown to assess its performance.

I. Introduction

Accurate estimation of the received signal angle of arrival (AOA) can be very beneficial for signal reception enhancement [1], [2] and wireless positioning [3], [4]. If the AOA estimates were not estimated accurately, then wireless devices
would would
will not be located accurately or the received signal bit error rate will be high.

Saleh O. Al-Jazzar is with the Electrical Engineering Department at the University of Hail, Hail, Saudi Arabia.

Thus, AOA estimators are developed and modified in such cases to increase their accuracy.

In this paper, we propose an AOA estimator that enhances the AOA estimation. The proposed method is named the Cholesky-decomposition-based-AOA (CDBA) estimator. The CDBA method utilizes the received signal at the two sides of the antenna array doublets. First, the autocorrelation matrix of the received signal at one side of the antenna array doublets is calculated. Then, the received signal autocorrelation matrix is decomposed using the Cholesky decomposition. A cross-correlation matrix between the received signals at both sides of the antenna doublets is calculated. The decomposed matrices of the autocorrelation matrix together with the crosscorrelation matrix are used to form a new matrix from which the AOA can be estimated with high accuracy.

In comparison with the well-known ESPRIT algorithm, the decomposition in the CDBA method is performed on a smaller matrix dimension. Also, the accuracy of the CDBA method is higher than that of the ESPRIT method.

The paper is organized as follows: Section II introduces the system model that forms the foundation for our estimator. The proposed CDBA estimator is presented in Section III. Section IV presents the simulated performance of the CDBA estimator. Finally, conclusions are shown in Section V.

II. System Model

In this section, we present the narrowband received signal model that will be utilized for the AOA estimation. The antenna array is formed from M uniform antenna doublets (i.e., $2M$ total antenna elements). Each antenna doublet is formed of two antenna elements spaced by a distance d. We assume a K BPSK sources signals, $\{\tilde{s}_k(t)\}_{k=1}^{K}$, impinging upon the antenna array. The signal $\tilde{s}_k(t)$ is represented as $\tilde{s}_k(t) = s_k(t)\cos(2\pi f_c t) = \Re\{s_k(t)\exp(j2\pi f_c t)\}$ where f_c is the carrier frequency and $s_k(t) = \sum_i \alpha_k(i) g(t - iT)$, i is an integer which represents the time index and $s_k(t)$ is the complex low-pass equivalent of $\tilde{s}_k(t)$, T is the

BPSK -> {-1,1}

symbol period, $\alpha_k(i) = A_k \beta_k(i)$ where $\beta_k(i) \in (-1, +1)$ which represents the symbol parity and A_k is a positive constant which represents the amplitude of $s_k(t)$ and $g(t)$ is the (raised cosine) pulse shaping function with

$$g(nT) = \left\{ \begin{array}{ll} 1 & \text{if } n = 0 \\ 0 & \text{otherwise} \end{array} \right\} \tag{1}$$

where n is an integer so that $s_k(nT) = \alpha_k(n)$.

The AOA is taken between the antenna array-axis and the source signal arrival direction and will be given the notation (θ_k). In this paper, we use the superscript notation $\{\hat{}\}$ to denote the estimated value of a variable (for example $\hat{h}$ is the estimated value of h). So, to start developing the received signal model, let us first consider the sampled received signal at the antenna elements (after the matched filter stage), from $m = 1$ to $m = M$, that are located in the first side of the antenna doublets, and let us call it $\mathbf{r}_1(i)$, and is represented as

$$\mathbf{r}_1(i) = \sum_{k=1}^{K} \alpha_k(i)\mathbf{a}(\theta_k) + \mathbf{n}_1(i) \tag{2}$$

where

$$\mathbf{a}(\theta_k) = \left[\, a_1(\theta_k) \quad \cdots \quad a_M(\theta_k) \,\right]^T \tag{3}$$

and

$$a_m(\theta_k) = \exp\{j\frac{2\pi}{\lambda}d \times (m-1)\cos(\theta_k)\} \tag{4}$$

with λ is the signal wavelength and d is the distance between consecutive antenna elements (the same as the distance between the elements of both sides of each antenna doublet). In addition, $\mathbf{n}_1(i)$ is the noise vector added to the received signal at the antenna elements (from $m = 1 \rightarrow M$) located along the first side of the antenna doublets, which is additive white Gaussian noise (AWGN) and has a covariance matrix of $\sigma^2\mathbf{I}_{M\times M}$, where $\mathbf{I}_{M\times M}$ is the $M \times M$ identity matrix. Finally,

$(.)^T$ represents the transpose operation.

The $(M \times 1)$ received signal vector at the first side of the antenna doublets set can be written in matrix form as

$$\mathbf{r}_1(i) = \mathbf{A}(\boldsymbol{\theta})\boldsymbol{\alpha}(i) + \mathbf{n}_1(i)$$

where

$$\mathbf{A}(\boldsymbol{\theta}) = [\,\mathbf{a}(\theta_1) \quad \cdots \quad \mathbf{a}(\theta_K)\,] \ , \tag{5}$$

$$\boldsymbol{\theta} = [\,\theta_1 \quad \cdots \quad \theta_K\,]^T$$

and

$$\boldsymbol{\alpha}(i) = [\,\alpha_1(i) \quad \cdots \quad \alpha_K(i)\,]^T \ .$$

From now on we will drop the term (i) from all terms for simplicity.

The received signal vector at the second side of the antenna doublets set will be given the notation $\mathbf{r}_2$. Thus,

$$\mathbf{r}_2 = \mathbf{A}(\boldsymbol{\theta})\mathbf{Z}\boldsymbol{\alpha} + \mathbf{n}_2$$

where

$$\mathbf{Z} = \mathrm{diag}\,[\,z_1 \quad \ldots \quad z_k \quad \cdots \quad z_K\,] \tag{6}$$

with

$$z_k = \exp\{j\frac{2\pi}{\lambda}d \times \cos(\theta_k)\} \tag{7}$$

and $\mathbf{n}_2$ is an AWGN vector at the second side of the antenna array doublets.

Now that we showed the received signal model at the antenna arrays, we will next propose the CDBA estimator.

III. Proposed Cholesky-Decomposition-Based AOA (CDBA) estimator

To implement the CDBA method we start by formulating an autocorrelation matrix of $\mathbf{r}_1$, as follows

$$\tilde{\mathbf{R}}_{11} = E[\mathbf{r}_1\mathbf{r}_1^H] = \mathbf{A}(\boldsymbol{\theta})\mathbf{P}_s\mathbf{A}(\boldsymbol{\theta})^H + \sigma^2\mathbf{I}_{M\times M} \tag{8}$$

where

$$\mathbf{P}_s := E[\boldsymbol{\alpha}\boldsymbol{\alpha}^H] = \mathrm{diag}\left(\begin{bmatrix} p_1 & \cdots & p_k & \cdots & p_K \end{bmatrix}\right)$$

where p_k is the power of the k^{th} source.

Now, if we take the Cholesky decomposition of $\tilde{\mathbf{R}}_{11}$, then

$$\tilde{\mathbf{R}}_{11} = \tilde{\mathbf{L}}\tilde{\mathbf{L}}^H \tag{9}$$

where $\tilde{\mathbf{L}}$ is a lower triangular matrix that results from the Cholesky decomposition.

Another matrix that we need to calculate is the crosscorrelation matrix ($\mathbf{R}_{21}$) between $\mathbf{r}_2$ and $\mathbf{r}_1$ as follows

$$\mathbf{R}_{21} = E[\mathbf{r}_2\mathbf{r}_1^H] = \mathbf{A}(\boldsymbol{\theta})\mathbf{P}_s\mathbf{Z}\mathbf{A}(\boldsymbol{\theta})^H \quad . \tag{10}$$

Now, forming a new matrix $\boldsymbol{\Psi}$ by multiplying the left and right hand sides of $\mathbf{R}_{21}$ by $\tilde{\mathbf{L}}^{-1}$ and $\tilde{\mathbf{L}}^{-H}$ as follows

$$\boldsymbol{\Psi} = \tilde{\mathbf{L}}^{-1}\mathbf{R}_{21}\tilde{\mathbf{L}}^{-H} \quad . \tag{11}$$

Then, the eigenvalue decomposition (EVD) for $\boldsymbol{\Psi}$ can be written as

$$\boldsymbol{\Psi} = \mathbf{U}\boldsymbol{\Gamma}_{\mathrm{Z}}\mathbf{U}^H \quad . \tag{12}$$

where $\boldsymbol{\Gamma}_{\mathrm{Z}}$ is the eigenvalue matrix of $\boldsymbol{\Psi}$ and $\mathbf{U}$ is the corresponding eigenvector matrix. The next subsections will show how the AOAs are estimated from the eigenvalue matrix $(\boldsymbol{\Gamma}_{\mathrm{Z}})$.

A. Verification for the CDBA estimator

Substituting (10) in (11) and comparing it with (12), then

$$\tilde{\mathbf{L}}^{-1}\mathbf{A}(\boldsymbol{\theta})\mathbf{P}_s\mathbf{Z}\mathbf{A}(\boldsymbol{\theta})^H\tilde{\mathbf{L}}^{-H} = \mathbf{U}\boldsymbol{\Gamma}_{\mathrm{Z}}\mathbf{U}^H \quad . \tag{13}$$

Defining the matrix $\mathbf{X}$ as follows

$$\mathbf{X} = \tilde{\mathbf{L}}^{-1}\mathbf{A}(\boldsymbol{\theta}) \quad . \tag{14}$$

Substituting (14) in (13), then

$$\mathbf{X}\mathbf{P}_s\mathbf{Z}\mathbf{X}^H = \mathbf{U}\mathbf{\Gamma}_{\mathrm{Z}}\mathbf{U}^H \quad . \tag{15}$$

Comparing both sides of (15)

$$\mathbf{X} = \mathbf{U}\mathbf{Q} \tag{16}$$

where $\mathbf{Q}$ is the appropriate matrix to change the basis.

Substituting (16) in (15), then

$$\mathbf{\Gamma}_{\mathrm{Z}} = \mathbf{Q}\mathbf{P}_s\mathbf{Z}\mathbf{Q}^H \quad . \tag{17}$$

The matrix $\mathbf{\Gamma}_{\mathrm{Z}}$ is an $M \times M$ matrix with $\mathbf{Z}$ and $\mathbf{P}_s$ are $K \times K$ diagonal matrices. Also, the nonzero diagonal elements of $\mathbf{\Gamma}_{\mathrm{Z}}$ are only the first K diagonal elements then (17) can be written as

$$\begin{bmatrix} \gamma_1 & \cdots & 0 & 0 & \cdots & 0 \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & \gamma_K & 0 & \cdots & 0 \\ 0 & \cdots & 0 & 0 & \cdots & 0 \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & 0 & \cdots & 0 \end{bmatrix}_{M \times M}$$

$$= \mathbf{Q} \begin{bmatrix} p_1 z_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & p_K z_K \end{bmatrix}_{K \times K} \mathbf{Q}^H \tag{18}$$

where the largest nonzero K values of the diagonal elements of $\mathbf{\Gamma}_Z$ correspond to the K sources which we will call $\gamma_{1\to K} \equiv \gamma_1 \to \gamma_K$. Each element of $\gamma_{1\to K}$ corresponds to one of the K sources.

Looking at (18), we can deduce that the matrix $\mathbf{Q}$ should be $M \times K$. Also, the lower $(M-K)\times K$ part of $\mathbf{Q}$ should be all zero elements, i.e.,

$$\mathbf{Q} = \begin{bmatrix} q_{1,1} & \cdots & q_{1,K} \\ \vdots & & \vdots \\ q_{K,1} & \cdots & q_{K,K} \\ 0 & \cdots & 0 \\ \vdots & & \vdots \\ 0 & \cdots & 0 \end{bmatrix}_{M\times K} = \begin{bmatrix} \mathbf{\Upsilon} \\ \mathbf{0} \end{bmatrix} \tag{19}$$

where

$$\mathbf{\Upsilon} = \begin{bmatrix} q_{1,1} & \cdots & q_{1,K} \\ \vdots & & \vdots \\ q_{K,1} & \cdots & q_{K,K} \end{bmatrix}_{K\times K} \tag{20}$$

and

$$\mathbf{0} = \begin{bmatrix} 0 & \cdots & 0 \\ \vdots & & \vdots \\ 0 & \cdots & 0 \end{bmatrix}_{(M-K)\times K} . \tag{21}$$

From (18) and the definition of $\boldsymbol{\Upsilon}$, $\mathbf{P}_s$ and $\mathbf{Z}$, we can deduce

$$\begin{bmatrix} \gamma_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \gamma_K \end{bmatrix}_{K\times K} = \boldsymbol{\Upsilon}\mathbf{P}_s\mathbf{Z}\boldsymbol{\Upsilon}^H \quad . \tag{22}$$

But, the left hand side of (22) is a diagonal matrix and the matrices $\mathbf{P}_s$ and $\mathbf{Z}$ are diagonal as well. So, from (22) we can deduce that the matrix $\boldsymbol{\Upsilon}$ is diagonal too, i.e.,

$$\boldsymbol{\Upsilon} = \begin{bmatrix} q_{1,1} & \cdots & 0 \\ \vdots & & \vdots \\ 0 & \cdots & q_{K,K} \end{bmatrix}_{K\times K} \quad . \tag{23}$$

Looking at the nonzero diagonal elements in both sides of (18), i.e., $\gamma_1 \rightarrow \gamma_K$, and from the definition of $\mathbf{Q}$ and $\boldsymbol{\Upsilon}$ in (19) and (23), respectively, the following

deduced

equation can be deuced:

$$\gamma_k = q_{k,k}q^*_{k,k}p_k z_k$$

where $(.)^*$ is the complex conjugate operator. Thus,

$$\gamma_k = |q_{k,k}|^2 p_k z_k \quad . \tag{24}$$

Since $|q_{k,k}|^2$ is a magnitude square and p_k is a power variable, then both terms are not complex variables. Thus, looking at both sides of (24), then

$$\angle\gamma_k = \angle z_k \quad . \tag{25}$$

B. Estimating the AOAs

The AOAs are implicated in $\angle z_k$ as shown in (7). Also, from (25) the phase of z_k equals the phase of γ_k which are the diagonal elements of $\mathbf{\Gamma}_{\mathrm{Z}}$. Thus, to find θ_k, we consider the k^{th} diagonal element of $\mathbf{\Gamma}_{\mathrm{Z}}$ as follows

$$\hat{\theta}_k = \cos^{-1}\left(\frac{\lambda\angle(\hat{\gamma}_k)}{2\pi d}\right) \tag{26}$$

where $\hat{\gamma}_k$ is the k^{th} diagonal element of the estimated $\hat{\mathbf{\Gamma}}_{\mathrm{Z}}$.

Thus, the required AOAs are estimated.

Note that the decomposition performed in third and fifth step is performed on a $M \times M$ matrix. Whereas, in the ESPRIT algorithm, the decomposition is will be performed on a $2M \times 2M$ matrix for the same system setup.

The performance of the proposed CDBA method will be shown in the simulation section (Section IV).

IV. Simulation Results

Simulations of the proposed CDBA estimator were completed to assess its performance. The elements of each antenna array were separated by a half-wavelength (i.e., $d = \frac{\lambda}{2}$). The number of sources was set to 2. A_1 was set to 1.2 and A_2 was set to 1. The proposed CDBA method was compared with the ESPRIT method.

Fig. 1 shows the root-mean-square-error (RMSE) of the AOA estimation in degrees for the proposed CDBA method compared with the ESPRIT method for different signal to noise ratios (SNRs). Since the power for both users were not equal, then we assumed the SNR was taken for the second user, i.e., if the SNR was set to 5dB then the second user has a 5dB SNR and the first user will have SNR of $20 \log 1.2 + 5 = 6.58$ dB. The angles for the two users were set to $\theta_1 = 30^o$ and $\theta_2 = 75^o$. The number of antenna doublets was set to $M = 3$. The number of

W=1000 not shown in Fig. 1

snap shots was set to $W = 200$ and $W = 1000$ (where W is the number of snap shots over which the correlation matrices were estimated). The results in Fig. 1 show that the proposed CDBA method gave better performance than the ESPRIT method. Also, the decomposition in the proposed CDBA method is performed on smaller matrices dimension than that in the ESPRIT method.

V. Conclusion

In this paper we propose an AOA estimator which we named the CDBA method. The CDBA method is applied by taking the Cholesky decomposition of the received signal autocorrelation matrix. The resultant decomposed matrices are used to modify the crosscorrelation matrix of the received signals at the antenna array doublets. The proposed CDBA method has better performance than the ESPRIT method in estimating the AOA. Also, the proposed method decomposes smaller matrices than the case in the ESPRIT algorithm. The performance of the proposed CDBA method was assessed and compared to the ESPRIT method.

REFERENCES

[1] N. Hew and N. Zein, "Space-time estimation techniques for UTRA system," *Capacity and Range Enhancement Techniques for the Third Generation Mobile Communications and Beyond (Ref. No. 2000/003), IEE Colloquium on*, pp. 6/1 – 6/7, February 2000.

[2] Y.-F. Chen and M. Zoltowski, "Joint angle and delay estimation for DS-CDMA with application to reduced dimension space-time RAKE receivers," *Acoustics, Speech, and Signal Processing, 1999. ICASSP '99. Proceedings., 1999 IEEE International Conference on*, vol. 5, pp. 2933 – 2936, March 1999.

[3] S. Al-Jazzar, M. Ghogho, and D. McLernon, "A joint TOA/AOA constrained minimization method for locating wireless devices in non-line-of-sight environment," *IEEE Transactions on Vehicular Technology*, vol. 59, pp. 468–472, January 2009.

[4] P. Deng and P. Fan, "An AOA assisted TOA positioning system," *Communication Technology Proceedings, 2000. WCC - ICCT 2000. International Conference on*, vol. 2, pp. 1501 – 1504, August 2000.

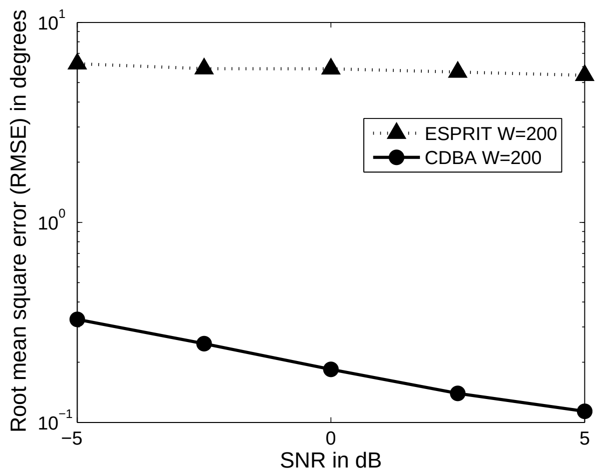

Fig. 1. Root mean square error (RMSE) of angle estimation in degrees versus signal to noise ratio (SNR) in dB for $(\theta_{1,2} = 30^o, 75^o)$.